WORKED SOLUTION

STEP II 2024, Question 11

Probability and Statistics • Group Testing Expectation

STEP II 2024, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Pooling only pays off if the group test usually comes back clear, and the expected number of tests per person turns out to be $1 + \frac{1}{k} - (1-p)^k$. Asking when this is at most 1 becomes a comparison between $(1-p)$ and $k^{-1/k}$, and that is exactly why part (i) asks about $x^{1/x}$: the best group size for a high p is the integer that makes $k^{1/k}$ largest.

Solution

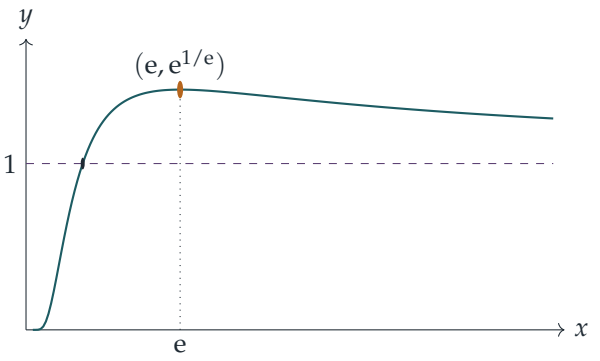
Part (i)

For $x > 0$ write $y = x^{1/x} = \mathrm{e}^{(\ln x)/x}$, so $\ln y = \dfrac{\ln x}{x}$. Differentiating,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1 - \ln x}{x^2}.$$

Since $y > 0$, $\frac{\mathrm{d}y}{\mathrm{d}x}$ has the sign of $1 - \ln x$: positive for $0 < x < \mathrm{e}$, zero at $x = \mathrm{e}$, negative for $x > \mathrm{e}$. So there is a single turning point, a maximum at $(\mathrm{e}, \mathrm{e}^{1/\mathrm{e}}) \approx (2.72,\ 1.44)$.

For the ends of the graph: as $x \to 0^+$, $(\ln x)/x \to -\infty$, so $y \to 0$; as $x \to \infty$, $(\ln x)/x \to 0$, so $y \to 1$ from above (for $x > 1$, $\ln y > 0$). The curve passes through $(1, 1)$.



For positive integers n, the function is increasing on $(0, \mathrm{e}]$ and decreasing on $[\mathrm{e}, \infty)$, so $1^{1/1} < 2^{1/2}$ and $3^{1/3} > 4^{1/4} > 5^{1/5} > \cdots$. The maximum is therefore $2^{1/2}$ or $3^{1/3}$. Raising both to the sixth power gives $2^3 = 8$ and $3^2 = 9$, so $3^{1/3} > 2^{1/2}$ and

$$\max_n n^{1/n} = 3^{1/3}.$$

Part (ii)

Write $q = 1 - p$. Consider one group of k people. By independence, the probability that nobody in the group has the enzyme is q^k, and only then is the pooled sample clear. So the group uses

$$\begin{cases} 1 \text{ test} & \text{with probability } q^k, \\ 1 + k \text{ tests} & \text{with probability } 1 - q^k, \end{cases}$$

and its expected number of tests is $1 + k(1 - q^k)$. There are $r = N/k$ groups, and expectation is additive, so the expected total is

$$E = \frac{N}{k}(1 + k(1 - q^k)) = N\left(1 + \frac{1}{k} - (1 - p)^k\right).$$

Part (iii)

From part (ii), $E \leq N$ if and only if

$$\frac{1}{k} \leq (1 - p)^k \iff 1 - p \geq k^{-1/k} = \frac{1}{k^{1/k}},$$

taking positive kth roots (both sides are positive, and $t \mapsto t^{1/k}$ is increasing).

? **Ask yourself** Which k allows the largest p?

For a given k the condition is $p \leq 1 - k^{-1/k}$, and this bound is largest when $k^{1/k}$ is largest. By part (i), $k^{1/k} \leq 3^{1/3}$ for every integer $k > 1$, so whatever k is chosen we need $p \leq 1 - 3^{-1/3}$. Conversely, N is a multiple of 3, so $k = 3$ is an allowed group size, and with $k = 3$ the condition is exactly $p \leq 1 - 3^{-1/3}$ (with equality giving $E = N$). Hence

the largest value is

$$p = 1 - 3^{-1/3} \approx 0.307.$$

To show this exceeds $\frac{1}{4}$:

$$1 - 3^{-1/3} > \tfrac{1}{4} \iff 3^{-1/3} < \tfrac{3}{4} \iff \tfrac{1}{3} < \tfrac{27}{64} \iff 64 < 81,$$

cubing in the middle step (cubing preserves order for positive numbers). The last statement is true, so $1 - 3^{-1/3} > \frac{1}{4}$.

Part (iv)

By the binomial expansion,

$$(1-p)^k = 1 - kp + \binom{k}{2}p^2 - \cdots.$$

If pk is small then $\binom{k}{2}p^2 < \frac{1}{2}(pk)^2$ is much smaller again, as are the later terms, so $(1-p)^k \approx 1 - pk$. Substituting into part (ii),

$$E \approx N\left(1 + \frac{1}{k} - (1 - pk)\right) = N\left(\frac{1}{k} + pk\right).$$

(In fact Bernoulli's inequality $(1-p)^k \geq 1 - pk$ shows the approximation never underestimates E, and stopping the expansion one term later shows the error is at most $\frac{1}{2}(pk)^2 N$.)

With $p = 0.01$ and $k = 10$ we have $pk = 0.1$, which is small, and

$$E \approx N\left(\frac{1}{10} + 0.1\right) = 0.2N,$$

so the scheme needs only about 20% of the N tests that testing everyone separately would use.

Checks

- Exactly, with $p = 0.01$ and $k = 10$: $E/N = 1.1 - 0.99^{10} \approx 1.1 - 0.9044 = 0.1956$, about 19.6%, within the error bound $\frac{1}{2}(0.1)^2 = 0.005$ of the approximation.
- At $p = 1 - 3^{-1/3}$ and $k = 3$: $(1-p)^3 = \frac{1}{3}$, so $E = N(1 + \frac{1}{3} - \frac{1}{3}) = N$, exactly on the boundary.
- Sanity of part (ii): as $p \to 0$, $E \to N/k$ (one test per group); as $p \to 1$, $E \to N(1 + \frac{1}{k})$ (every group is retested). The formula was also confirmed by computer, by enumerating all outcomes of small groups.

Common slips

- Answering part (i) with $e^{1/e}$: n must be an integer, so compare $2^{1/2}$ and $3^{1/3}$.

- In part (iii), checking only $k = 3$ without explaining why no other group size could allow a larger p.
- Forgetting the pooled test itself in a positive group, giving k rather than $k + 1$ tests.

Beyond STEP

This is Dorfman's group testing scheme, devised in the Second World War to screen soldiers' blood samples, and it was revived for COVID-19 testing. Minimising $\frac{1}{k} + pk$ gives $k \approx 1/\sqrt{p}$ (so $k = 10$ is optimal here) and about $2\sqrt{p}\,N$ tests; smarter adaptive schemes, studied in information theory, do better still. For the probability behind it, see Blitzstein and Hwang, *Introduction to Probability*.

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